3D Animated Solar System

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  background: #02030b;
  color: white;
  font-family: Arial, sans-serif;
  overflow: hidden;
}

h1 {
  text-align: center;
  padding: 14px 5px;
  font-size: 24px;
  position: relative;
  z-index: 2;
  background: rgba(0,0,0,0.35);
}

#scene {
  width: 100%;
  height: calc(100vh - 65px);
```

```
  display: block;
}

#status {
  position: absolute;
  top: 65px;
  left: 10px;
  z-index: 3;
  font-size: 12px;
  color: #9eeaff;
}

.controls {
  position: absolute;
  bottom: 18px;
  left: 0;
  right: 0;
  z-index: 3;
  text-align: center;
}

button {
  background: #123c68;
  color: white;
  border: 1px solid #5bcaff;
  border-radius: 20px;
  padding: 10px 16px;
  margin: 4px;
  font-size: 14px;
}
```

Ø<ß 3 D S O L A R S Y S T E M

Loading 3D Solar System...

```
  Pause / Play
  Reset View



window.addEventListener("load", function() {

  const status = document.getElementById("status");

  if (typeof THREE === "undefined") {
    status.textContent =
      "Three.js load failed. Check internet or CDN access.";
    return;
  }

  if (typeof THREE.OrbitControls === "undefined") {
    status.textContent =
      "OrbitControls load failed. Check internet connection.";
    return;
  }

  status.textContent = "3D Solar System Loaded Successfully!";

  const container = document.getElementById("scene");

  const scene = new THREE.Scene();
  scene.background = new THREE.Color(0x02030b);

  const camera = new THREE.PerspectiveCamera(
    60,
    container.clientWidth / container.clientHeight,
    0.1,
    1000
  );
```

```
camera.position.set(0, 35, 65);

const renderer = new THREE.WebGLRenderer({
  antialias: true
});

renderer.setSize(
  container.clientWidth,
  container.clientHeight
);

renderer.setPixelRatio(
  Math.min(window.devicePixelRatio, 2)
);

container.appendChild(renderer.domElement);

const controls = new THREE.OrbitControls(
  camera,
  renderer.domElement
);

controls.enableDamping = true;
controls.minDistance = 12;
controls.maxDistance = 130;

// Lighting
const ambient = new THREE.AmbientLight(0xffffff, 0.35);
scene.add(ambient);

const sunLight = new THREE.PointLight(
  0xffffff, 2.5, 300
);
scene.add(sunLight);

// Stars
const starGeometry = new THREE.BufferGeometry();
const starPositions = [];
```

```
for (let i = 0; i < 1800; i++) {
  starPositions.push(
    (Math.random() - 0.5) * 250,
    (Math.random() - 0.5) * 250,
    (Math.random() - 0.5) * 250
  );
}

starGeometry.setAttribute(
  "position",
  new THREE.Float32BufferAttribute(starPositions, 3)
);

const stars = new THREE.Points(
  starGeometry,
  new THREE.PointsMaterial({
    color: 0xffffff,
    size: 0.35
  })
);

scene.add(stars);

// Sun
const sunGeometry = new THREE.SphereGeometry(4, 40, 40);

const sunMaterial = new THREE.MeshBasicMaterial({
  color: 0xffaa00
});

const sun = new THREE.Mesh(sunGeometry, sunMaterial);
scene.add(sun);

// Sun glow
const glow = new THREE.Mesh(
  new THREE.SphereGeometry(4.8, 32, 32),
  new THREE.MeshBasicMaterial({
    color: 0xff6600,
```

```
    transparent: true,
    opacity: 0.16
  })
);

scene.add(glow);

// Planet information
const planetData = [
  { name: "Mercury", size: 0.55, distance: 7, color: 0xaaaaaa, speed: 0.025 },
  { name: "Venus", size: 0.9, distance: 10, color: 0xffbb66, speed: 0.018 },
  { name: "Earth", size: 1, distance: 14, color: 0x2288ff, speed: 0.014 },
  { name: "Mars", size: 0.75, distance: 18, color: 0xff4422, speed: 0.012 },
  { name: "Jupiter", size: 2.5, distance: 25, color: 0xd6a16c, speed: 0.008 },
  { name: "Saturn", size: 2.1, distance: 33, color: 0xe6cc88, speed: 0.006 },
  { name: "Uranus", size: 1.5, distance: 41, color: 0x66ddff, speed: 0.004 },
  { name: "Neptune", size: 1.45, distance: 48, color: 0x3355ff, speed: 0.003 }
];

const planets = [];

planetData.forEach(function(data) {

  // Orbit line
  const orbitGeometry = new THREE.RingGeometry(
    data.distance - 0.025,
    data.distance + 0.025,
    120
  );

  const orbitMaterial = new THREE.MeshBasicMaterial({
    color: 0x445577,
    side: THREE.DoubleSide,
    transparent: true,
    opacity: 0.6
  });

  const orbit = new THREE.Mesh(
```

```
  orbitGeometry,
  orbitMaterial
);

orbit.rotation.x = Math.PI / 2;
scene.add(orbit);

// Planet pivot
const pivot = new THREE.Object3D();
scene.add(pivot);

const geometry = new THREE.SphereGeometry(
  data.size, 32, 32
);

const material = new THREE.MeshStandardMaterial({
  color: data.color,
  roughness: 0.8
});

const planet = new THREE.Mesh(geometry, material);

planet.position.x = data.distance;
pivot.add(planet);

// Saturn rings
if (data.name === "Saturn") {
  const ring = new THREE.Mesh(
    new THREE.RingGeometry(2.7, 4, 48),
    new THREE.MeshBasicMaterial({
      color: 0xc9ae82,
      side: THREE.DoubleSide
    })
  );

  ring.rotation.x = Math.PI / 2.5;
  planet.add(ring);
}
```

```
  // Moon around Earth
  if (data.name === "Earth") {
    const moonPivot = new THREE.Object3D();
    planet.add(moonPivot);

    const moon = new THREE.Mesh(
      new THREE.SphereGeometry(0.27, 20, 20),
      new THREE.MeshStandardMaterial({
        color: 0xdddddd
      })
    );

    moon.position.x = 1.8;
    moonPivot.add(moon);

    planet.userData.moonPivot = moonPivot;
  }

  planets.push({
    pivot: pivot,
    planet: planet,
    speed: data.speed
  });
});

let running = true;

window.toggleAnimation = function() {
  running = !running;
};

window.resetView = function() {
  camera.position.set(0, 35, 65);
  controls.target.set(0, 0, 0);
  controls.update();
};

// Animation
```

```
  function animate() {
    requestAnimationFrame(animate);

    if (running) {
      planets.forEach(function(item) {
        item.pivot.rotation.y += item.speed;
        item.planet.rotation.y += 0.01;

        if (item.planet.userData.moonPivot) {
          item.planet.userData.moonPivot.rotation.y += 0.04;
        }
      });

      sun.rotation.y += 0.002;
      glow.rotation.y -= 0.001;
    }

    controls.update();
    renderer.render(scene, camera);
  }

  animate();

  // Responsive screen
  window.addEventListener("resize", function() {
    const width = container.clientWidth;
    const height = container.clientHeight;

    camera.aspect = width / height;
    camera.updateProjectionMatrix();

    renderer.setSize(width, height);
  });

});
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